"Pre-Registered Protocol — DarkSide-20k Screening Comparison v1.0"

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Pre-Registered Protocol
Var-Rho Certification Screening vs the DarkSide-20k Acceptance Criteria: Phase-1 Simulation Study

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Companions: concept note v1.0 and exploratory skeleton v0.1 (seed 684), archived with this record. Wherever a skeleton number appears

below, it is an anchor — it motivated the lock, and it is not evidence for it.

Abstract
DarkSide-20k screens roughly a quarter-million silicon photomultiplier components into 648 photo-detector units, and it has published how it decides: a tile lives if its signal-to-noise clears 8, if its pulse-count tail stays under 1.8 Hz/mm², if its breakdown voltage sits within a volt of nominal. The program publishes its yield — 87% — and its correlated-noise parameters: k_dup = 0.445 ± 0.006 from the Proto-0 array, cryogenic characterization reporting fewer than 0.06 correlated delayed avalanches per pulse. This protocol pre-registers the study that asks the natural next question: does the Var-Rho statistics certification regime agree with that one? Not a claim that Var-Rho is better — a side-by-side, on identical simulated populations,

under criteria frozen before the first full-study tile exists. An exploratory skeleton run taught three lessons that now stand as rules: agreement must be judged only where both regimes can see (breakdown voltage is not a waveform observable); the correlated-noise band must be anchored to the collaboration's own published numbers, not to population quantiles; and no population is admissible until the published criteria reproduce the published yield. Six falsifiers, D1–D6, carry numeric criteria, and every reported rate carries a Wilson 95% interval or it does not get reported.

Keywords: SiPM; quality assurance; photon detector; dark matter; correlated noise; simulation; pre-registration; Var-Rho statistics certification; var-rho; statistics-certification

1. Why this study exists

   The concept note poses the question; the

skeleton (seed 684, archived alongside) explored it with uncalibrated populations and population-quantile bands, and what it found is worth retelling, because three of its numbers became three rules.

The first was 73.9% — the raw keep/reject agreement between the two regimes. Low, until you ask where they disagree: breakdown-voltage outliers, which a pulse-stream regime is blind to by construction. A gate that reads waveforms cannot see an electrical parameter, and a study that counts its blindness as disagreement would be grading a fish on climbing. Rule 1 follows: D6 measures agreement on shared-observable classes only — SNR failures, PCR failures, correlated failures, good tiles — and the breakdown blind region is quantified separately, in the open.

The second was 27.9% — the fraction of

injected high-correlated tiles the skeleton’s quantile band caught. The band sat where the accepted population’s upper tail sat, and the bad population lived inside that tail. Rule 2 follows: the correlated band is anchored to the published measurements themselves, CDA below 0.06 per pulse and k_dup = 0.445 ± 0.006, mapped once and honestly onto the simulation’s correlated-fraction scale, with the mapping recorded and stress-tested by a ±25% sensitivity arm. The collaboration measured the anchor; this study borrows it. The third was 62% — the yield the skeleton’s population produced under the published cuts, against the published 87%. The breakdown spread was too wide, and the criteria were rejecting a quarter of the population before anyone’s screening philosophy entered the room. Rule 3 follows, and it gates everything: population spreads are calibrated until the

published criteria reproduce the published yield within a locked tolerance. If that gate cannot be met, the study’s finding is the mismatch itself, reported and closed. A simulator that cannot reproduce the reference numbers has no business comparing screening regimes.

2. Definitions

   Population model: per-tile SNR, pulse-count rate, breakdown offset, and correlated fraction, drawn from calibrated spreads, with injected failure classes (low SNR, high PCR, breakdown outliers, high correlated) at fractions the yield-reproduction gate itself fixes.

   Regime DS: the published acceptance criteria, applied exactly as published — SNR above 8, PCR at or under 1.8 Hz/mm², breakdown within ±1 V.

Regime VR: the Var-Rho certification bands. SNR from the first percentile of the DS-accepted population upward; PCR from zero to the 99th percentile of the DS-accepted; correlated from zero to the Rule-2 anchor C. Bands are computed once, on a 20% calibration subset, and frozen.
Shared-observable classes: the classes both regimes can in principle decide — SNR failure, PCR failure, correlated failure, good. Breakdown is not among them, and the protocol does not pretend otherwise.
Certified reporting: a Wilson 95% score interval on every rate. A point estimate standing alone is not a reportable result in this study.

3. How the study runs

Calibration comes first and twice. On the 20% calibration subset the VR bands are set under

Rule 2, and the yield-reproduction gate of Rule 3 is verified on that same subset — spreads admitted only if the published cuts reproduce the published yield within tolerance. Any change to the spreads re-verifies the gate; the gate outranks everything downstream.
Rate-honesty runs on per-tile pulse-count streams, Poisson at the tile's PCR, under the machinery the IceCube study established and archived: inter-arrival KS tests against the dead-time-corrected null at the nominal operating rate, Fano factors in locked time bins, distortion corrected under the non-paralyzable relation with a hold-off in the 10 μs class, re-derived per regime and recorded rather than assumed.

4. Falsifiers
   Failure voids the claim; deviation is reported, never repaired. The criteria are fixed here,

before the full-study population exists.

ID

Claim protected

Criterion (fixed)

D1

Correlated-noise identification

≥ 95% identification (Wilson lower bound ≥ 92%) of injected high-correlated tiles, CDA-equivalent above the Rule-2 anchor

D2

Retention of accepted tiles

≥ 90% (lower bound ≥ 88%) retention of DS-accepted tiles under VR

D3

Rate-honesty

KS $p \geq 0.05$ against the dead-time-corrected null at the nominal regime; Fano within [0.9, 1.1]

D4

Threshold portability

Frozen bands across batch shifts of ±0.5 in population means: shared-class agreement within 3 points of the calibration-batch value

D5

No bias

VR efficiency flat across SNR and gain bins within ±5 points — breakdown bins excluded, not being a waveform observable

D6

Shared-observable agreement

Keep/reject agreement with DS on shared-observable classes ≥ 95% (lower bound ≥ 93%); the breakdown blind region quantified separately, with intervals

For calibration of expectation, the skeleton’s exploratory numbers sit in the record: overall agreement 73.9% [73.3–74.5], the figure Rule 1 superseded; retention 97.06% [96.7–97.3]; batch-to-batch agreement stable within two points; the 27.9% identification and 62% yield that Rules 2 and 3 exist to fix. They are the map’s warning labels, not its terrain.

5. What this study does not claim

Four limitations are stated before any result exists. VR certifies pulse-stream statistics; it does not replace electrical screening, and the blind region that creates is quantified, never hidden. The Rule-2 anchor crosses one recorded mapping from published avalanche parameters to the simulation’s correlated-fraction scale; the sensitivity arm exists because that mapping is a choice, and the study says which one it made. Agreement is measured against a published, idealized description of production QA; the real

screening carries detail this desk cannot see, and the results paper must mark every comparison that touches that gap. And this is an engineering analogue throughout — no statement here is about real DarkSide-20k tiles, and none will be without the collaboration's streams.

6. Versioning and deviation policy

Immutable once timestamped. Amendments are new versions with dated changelogs, issued before the runs they touch. Deviations are logged and disclosed with their statistical consequences; selective deviation voids pre-registration for the affected claim.

7. Lock-time parameter table

Parameter Source / value

1: Population spread calibration (precondition) ☐ (must pass Rule 3 gate)

2: Rule-2 mapping: CDA/k_dup → simulated correlated fraction ☐ (single recorded choice)

3: Calibration/test split realization☐ (seed 684 lineage)

4: Injected failure-class fractions☐ (fixed by Rule 3 gate)

5:Nominal operating rate for D3 ☐

6: Hold-off, per regime ☐ (10 μs class, re-derived)

7: Fano bin width ☐

Declarations

Funding: None (independent research).

Competing interests: None. Data and code availability: No data accompany this protocol. The exploratory skeleton notebook and the full-

study notebook — Run-All, seed-pooled — together with results and figures, will be archived with this record. Author contributions: N.K. designed the study, ran the exploratory skeleton, and wrote this protocol.

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Changelog

Version: 1.0

Date: 2026-10-20

Change:

Initial pre-registration, encoding the skeleton's three lessons as Rules 1–3 and fixing D1–D6.

References

DarkSide Collaboration, veto SiPM tile construction and characterization — SNR > 8, PCR tail 1.8 Hz/mm², breakdown ±1 V, 87% yield. [exact citation at study submission]

DarkSide Collaboration, Proto-0 array (2026) — k_dup = 0.445 ± 0.006. [exact citation at study

submission]
DarkSide Collaboration, arXiv:2610.18223 (October 2026) — cryogenic VUV4 characterization, CDA < 0.06 per pulse.
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N. Khan, QCID-Sigma v5.4, Zenodo 10.5281/zenodo.23378709 — hardware-verified baseline.
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